***COURSE*: CHE 312: Transport Phenomena II**

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

<u>Midsemester Exam #4</u>

Online

***Duration: 50 minutes --- May 1*, 2020**

Problem 1: (60 points)

If we have a single porous catalyst particle reactor in which the surface reaction rate is zeroth order with respect to the reactant species A in the chemical reaction A → B, determine:

(i) **1,2,3 (20 points)** the concentration profile of the species A in the particle, if the particle is spherical and isothermal, and $C_A(r = R) = C_{As}$

(ii) **1,2,3 (20 points)** the effectiveness factor as a function of the various system parameters, if the reactor temperature remains constant

(iii) **4 (10 points)** the conditions under which the porous catalyst particle reactor can be used for kinetic studies of the reaction A → B

(iv) **4 (10 points)** the conditions at which diffusion is the overall rate limiting process.

(Bonus 20 points) Repeat the calculation (i) for the case that

$$-D \cdot dC_A/dr = h_m \cdot (C_A - C_{Ab}) \text{ at } r = R,$$

where h_m is mass transfer coefficient and C_{Ab} is bulk known concentration of A.

Problem 2: (40 points)

Water enters a tube at 27 °C with a flow rate of 450 kg/h. The heat transfer from the tube wall to the fluid is given as q' (W/m) = $\alpha \cdot x$, where the coefficient α is 20 W/m^2 and x(m) is the axial distance from the tube entrance. Beginning with a properly defined differential control volume in the tube, derive an expression for the temperature distribution $T_m(x)$ of the water. What is the outlet temperature of the water for a heated section 30 m long? (Hint: for water, C_P = 4.18 kJ/(kg·K).